Key policies and mechanisms which help prevent or reduce congestion in data layer

1. Retransmission Policy: Determines when and which lost or corrupted frames should be retransmitted. Efficient retransmission mechanisms, such as Selective Repeat ARQ, retransmit only missing or damaged frames instead of all subsequent frames, reducing unnecessary traffic, bandwidth consumption, and congestion.

2. Out-of-Order Caching Policy (Selective Repeat): Allows the receiver to temporarily store correctly received frames that arrive out of order while waiting for missing frames. Once the missing frames arrive, the receiver delivers them in the correct sequence. This avoids unnecessary retransmissions, improves bandwidth efficiency, and reduces congestion, but requires additional buffer memory.

3. Acknowledgment Policy (Piggybacking): Combines acknowledgment (ACK) information with outgoing data frames in bidirectional communication instead of sending separate ACK frames. This reduces control-frame overhead, saves bandwidth, and helps prevent congestion. However, excessive ACK delays may cause unnecessary retransmissions, so acknowledgment timers are required.